



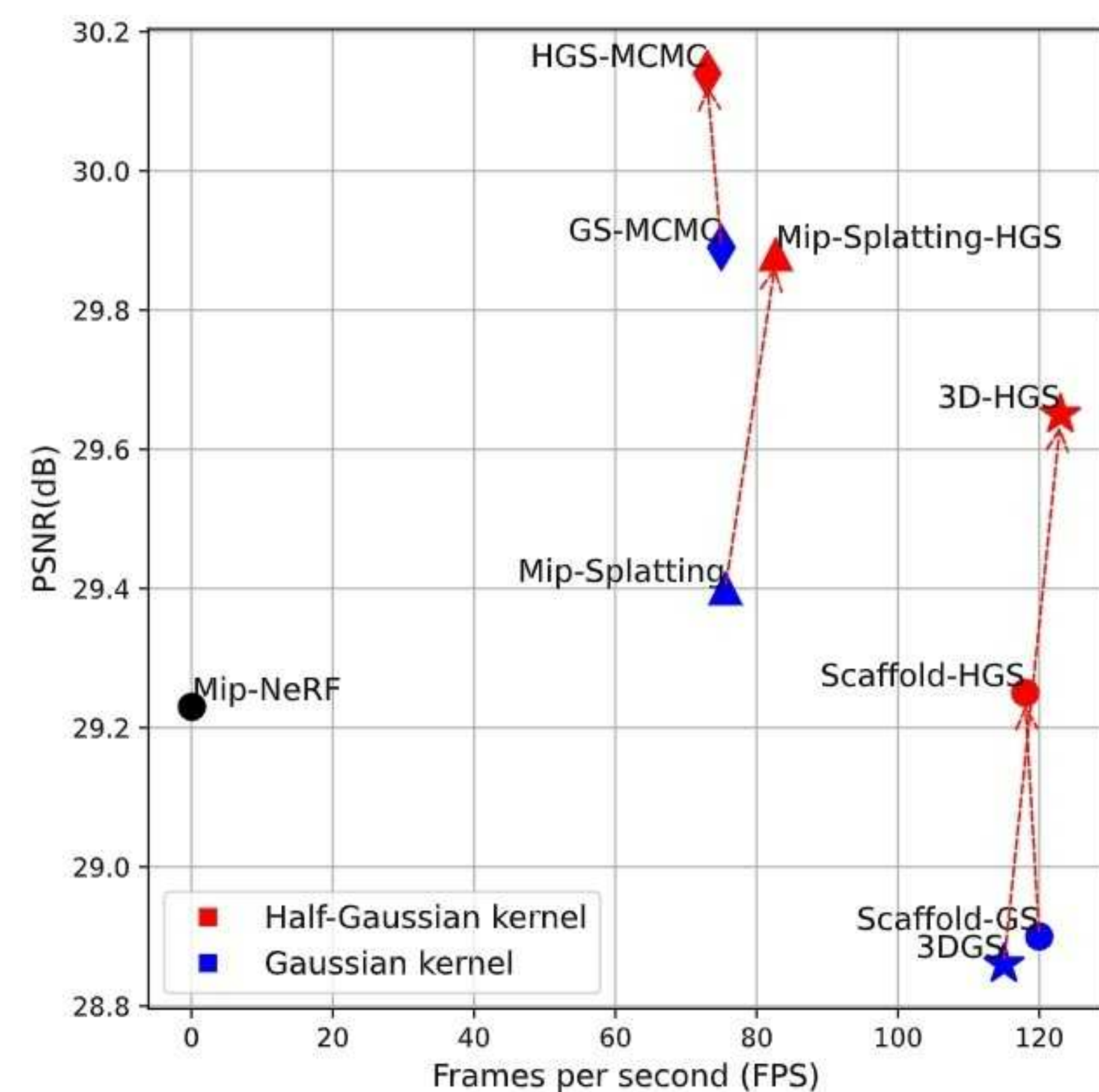
3D-HGS: 3D half gaussian splatting

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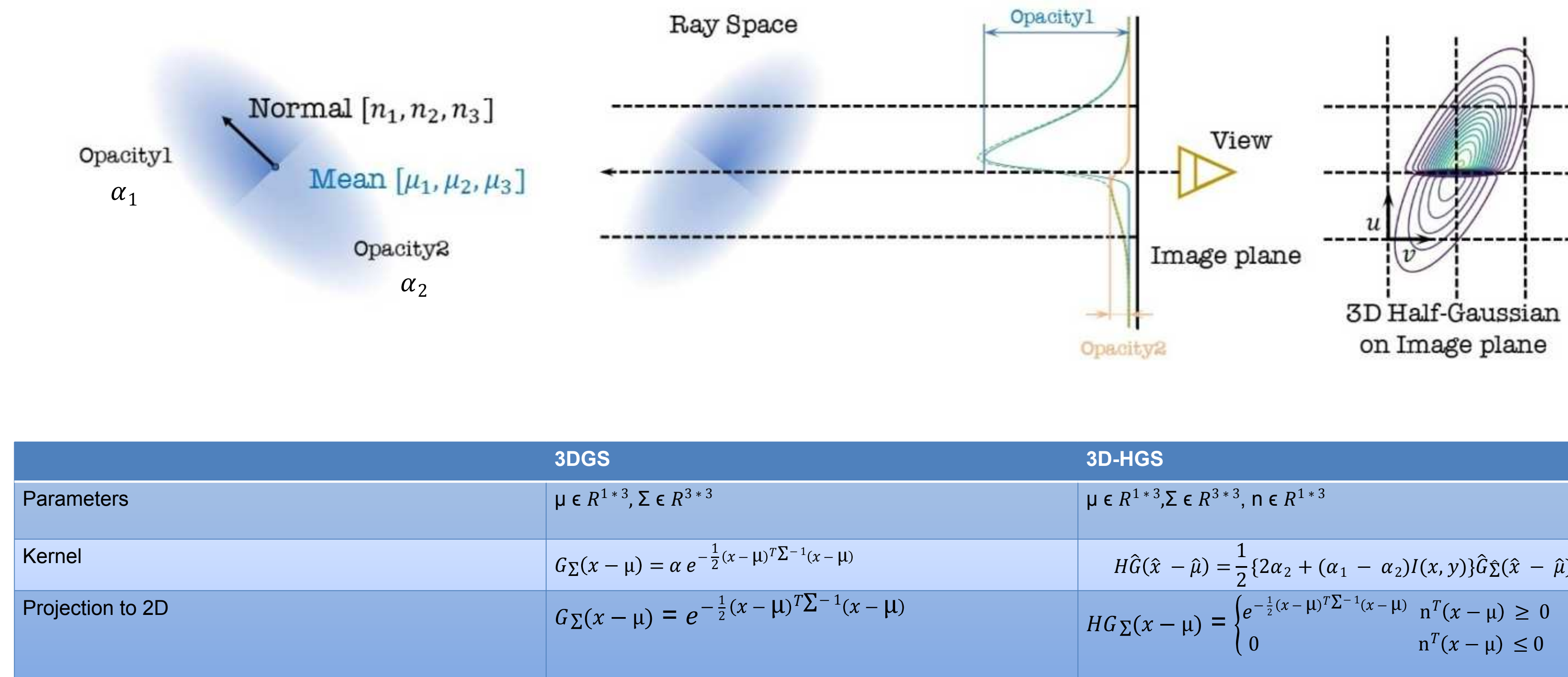


Motivation

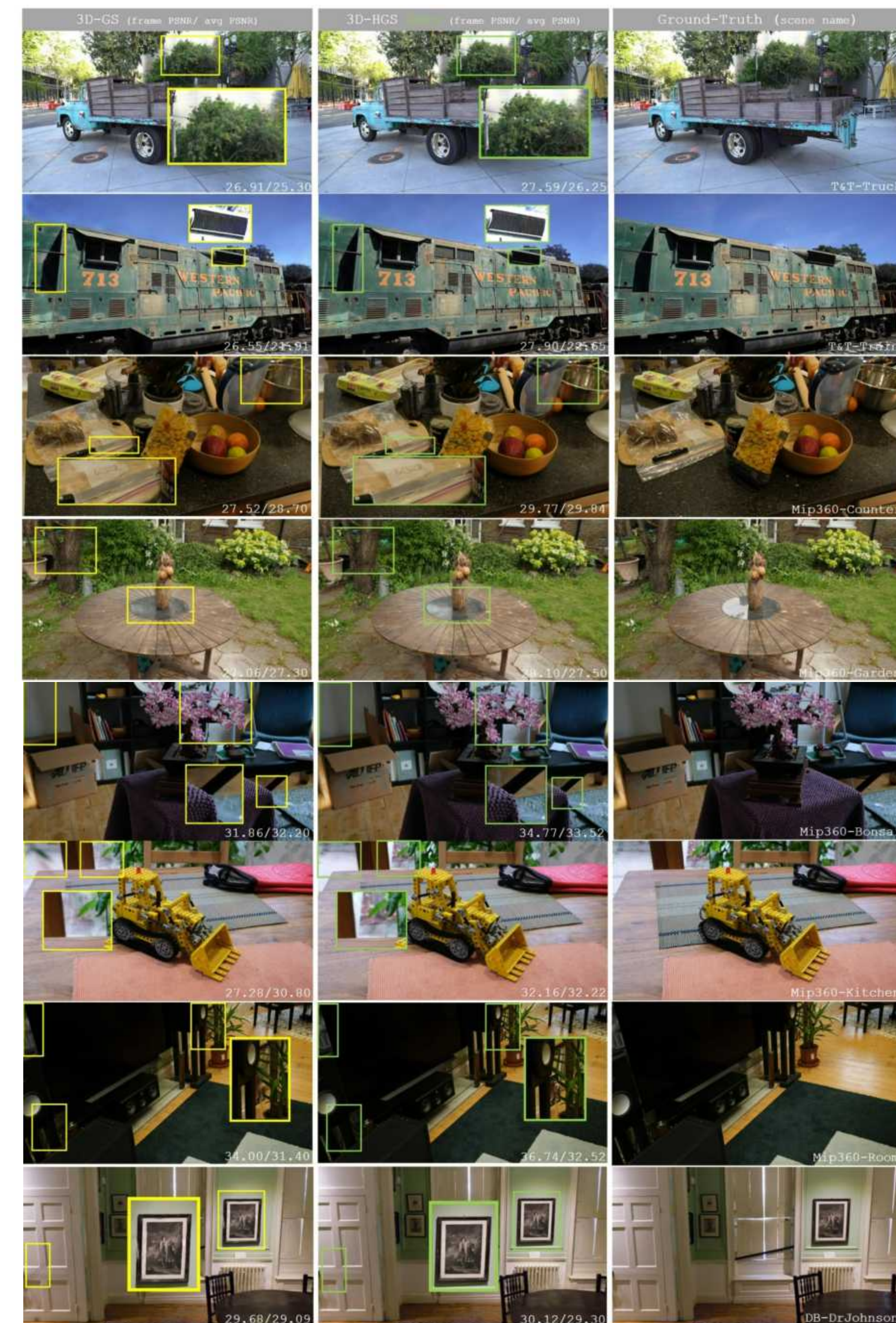


- **3D half gaussian splatting:**
- Introduces a cutting plane to enhance shape and color boundary representation.
 - Improves modeling of discontinuities in 3D Gaussian Splatting.
 - Designed as a plug-and-play kernel, compatible with other Gaussian Splatting methods.

Our Method: 3D-HGS

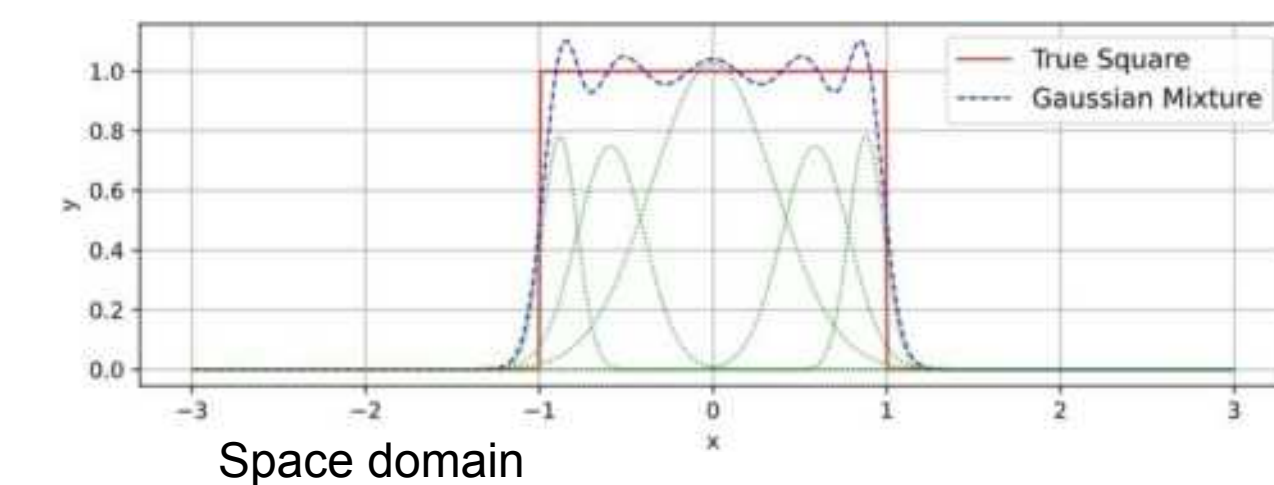


Qualitative Results

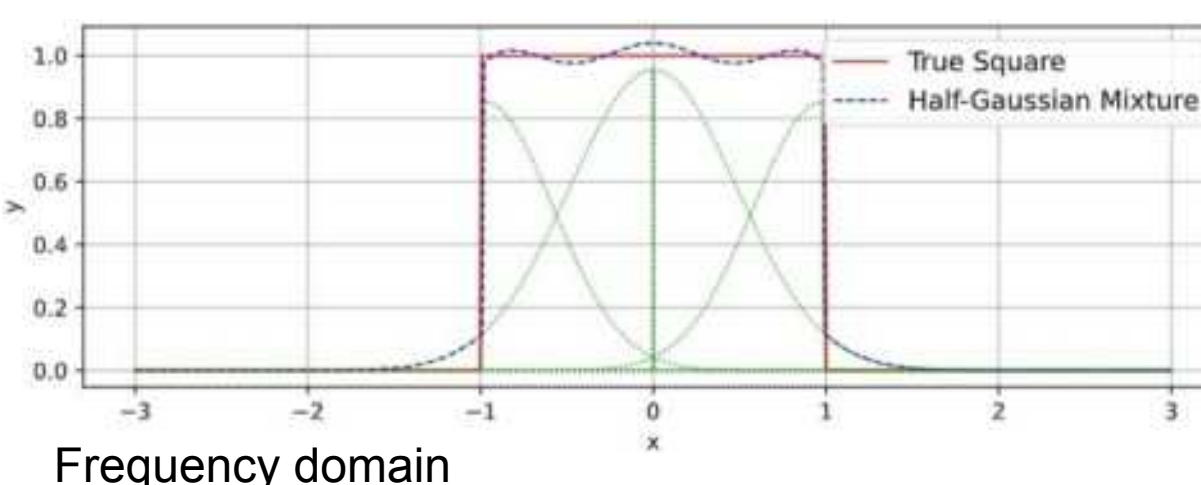


Comparison of Half-Gaussian and Gaussian Kernels

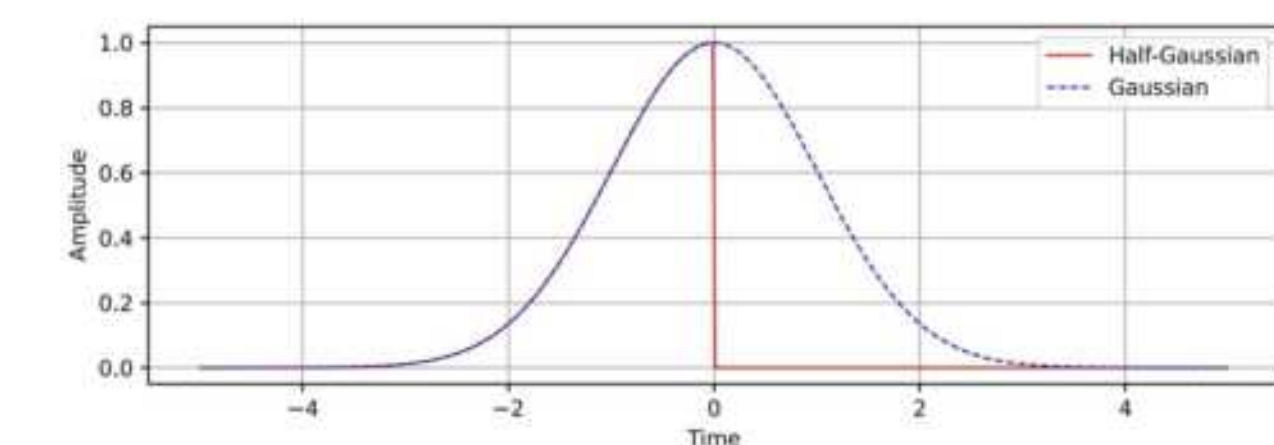
Five Gaussians fitting a square



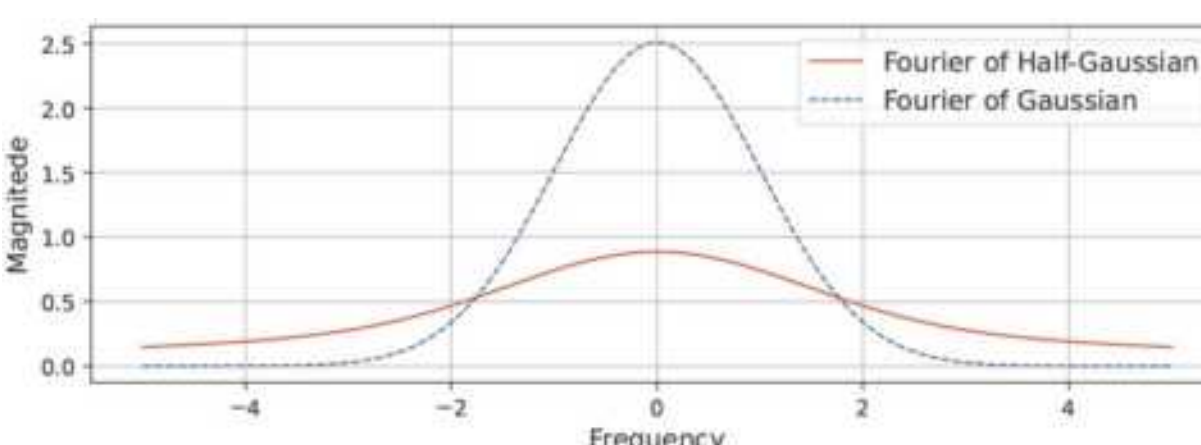
Three HGs fitting a square



Space domain



Frequency domain



Time Domain: Half-Gaussians better approximate sharp edges with fewer kernels and lower error.

Frequency Domain: Preserve more high-frequency components, improving edge fidelity.

Quantitative Results

Dataset Method	Mip-NeRF360			Tanks&Temples			Deep Blending		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Mip-NeRF [1]	29.23	0.844	0.207	22.22	0.759	0.257	29.40	0.901	0.245
2D-GS[8]	28.98	0.867	0.185	23.43	0.845	0.181	29.70	0.902	0.250
Fre-GS [30]	27.85	0.826	0.209	23.96	0.841	0.183	29.93	0.904	0.240
GES [6]	28.69	0.857	0.206	23.35	0.836	0.198	29.68	0.901	0.252
3D-GS [9]	28.88	0.870	0.182	23.60	0.847	0.181	29.41	0.903	0.243
3D-HGS (Ours)	29.66 +0.78	0.873	0.178	24.45 +0.85	0.857	0.169	29.76 +0.35	0.905	0.242
Scaffold-GS [15]	28.95	0.848	0.220	23.96	0.853	0.177	30.21	0.906	0.254
Scaffold-HGS(Ours)	29.25 +0.30	0.867	0.186	24.42 +0.46	0.859	0.162	30.36 +0.15	0.910	0.240
Mip-Splatting [29]	29.39	0.880	0.162	23.75	0.857	0.157	29.46	0.903	0.243
Mip-Splatting-HGS(Ours)	29.88 +0.49	0.881	0.160	24.53 +0.78	0.865	0.145	29.61 +0.15	0.901	0.241
GS-MCMC [10]	29.89	0.900	0.190	24.29	0.860	0.190	29.67	0.890	0.320
HGS-MCMC(Ours)	30.13 +0.24	0.886	0.158	25.08 +0.77	0.841	0.144	29.80 +0.13	0.898	0.245

$\int_{n^T(x-\mu)\geq 0}HG_{\Sigma}(x-\mu)dz=\frac{1}{2}I(x,y)\hat{G}_{\Sigma}(\hat{x}-\hat{\mu})$	
Parameters for half gaussian Gaussian $\mu \in R^{1*3}$ $\Sigma \in R^{3*3}$	The definition of Half gaussian kernel
	Compare GS and HGS
	$HG_{\Sigma}(x-\mu)=\begin{cases}e^{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)} & n^T(x-\mu)\geq 0 \\ 0 & n^T(x-\mu)\leq 0\end{cases}$
Cutting plane $n \in R^{1*3}$	$I(x,y)=erfc\left(-\frac{\frac{n_1x+n_2y}{n_3}+\mu_{z xy}}{\sqrt{2}\sigma_{z xy}}\right)$

The image C can be represented as

$$C=\sum_{i\in N}c_iH\hat{G}_i(\hat{x}-\hat{\mu})\prod_{j=1}^{i-1}(1-H\hat{G}_j(\hat{x}-\hat{\mu}))$$

α_1 : The opacity of the first half

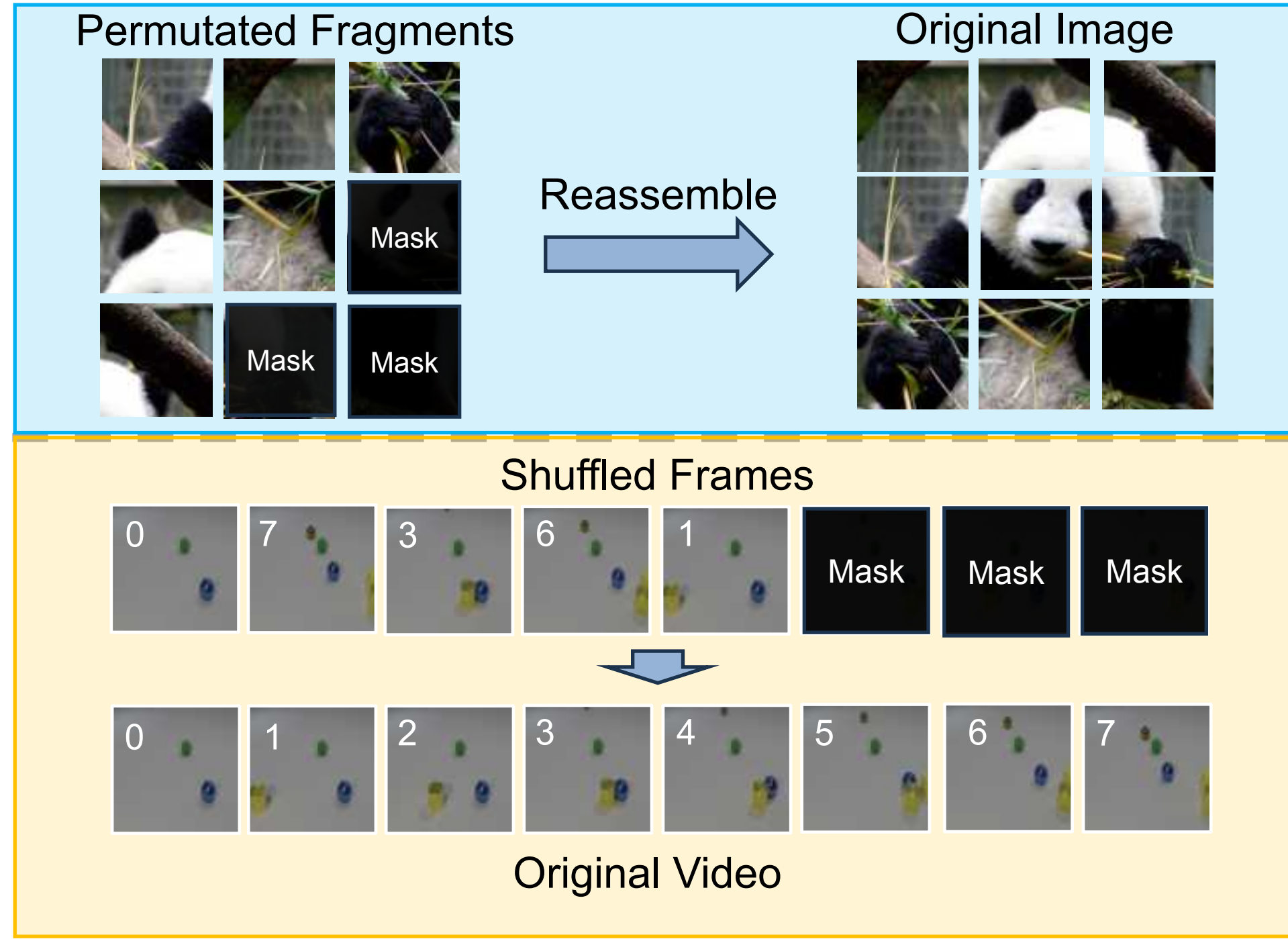
α_2 : The opacity for another half

$$H\hat{G}(\hat{x}-\hat{\mu})=\frac{1}{2}\{2\alpha_2+(\alpha_1-\alpha_2)I(x,y)\}\hat{G}_{\Sigma}(\hat{x}-\hat{\mu})$$

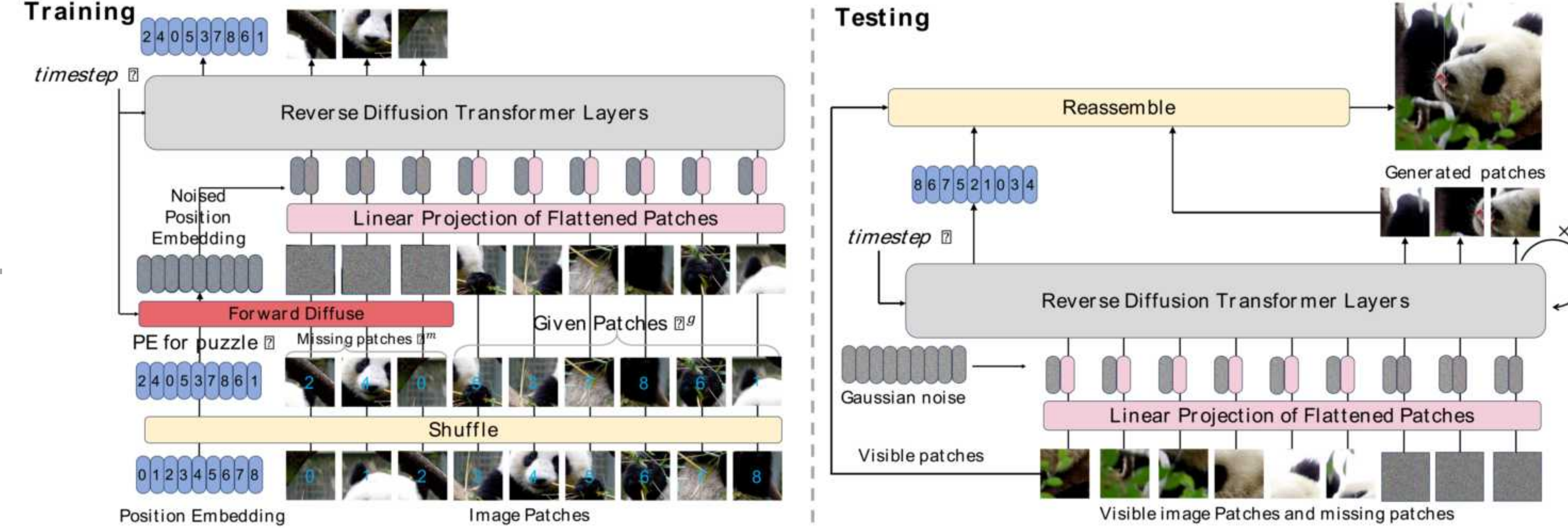
Masked Jigsaw Puzzles

Spatial

Temporal



Our Method: JPDVT



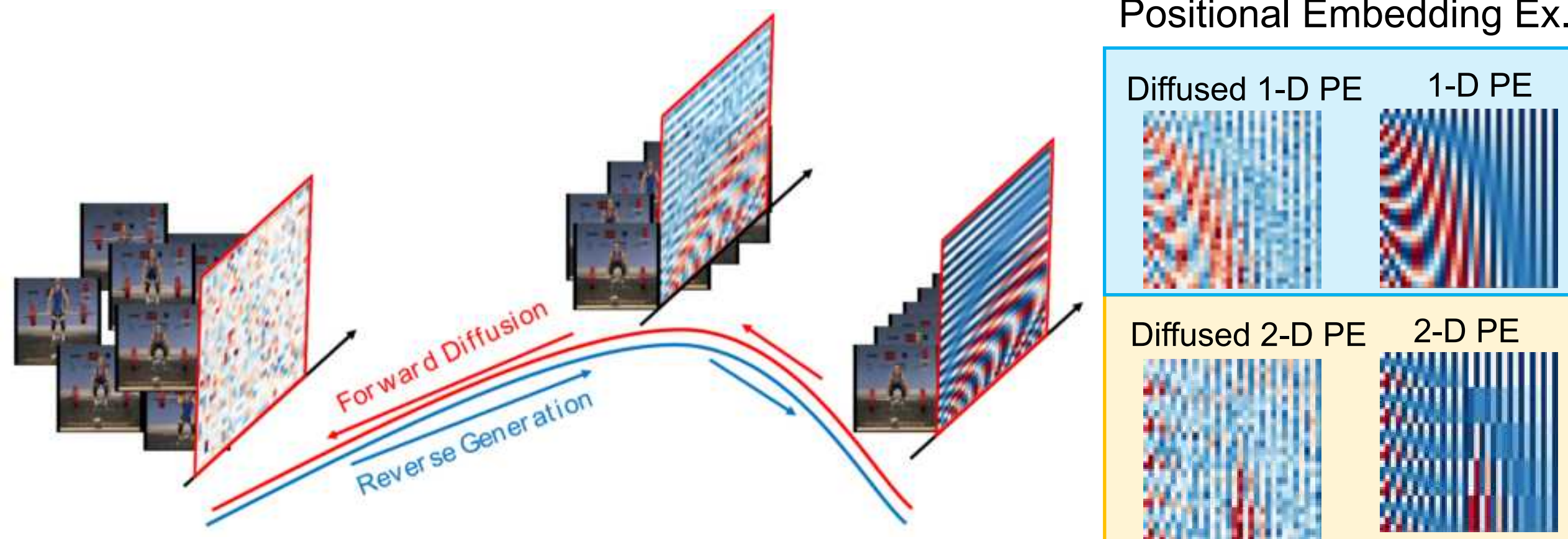
$E = \{e_1, e_2, \dots, e_N\} = E^g \cup E^m$ refers to puzzle pieces of both **given** and **missing**
 $L = (\gamma(l_1), \gamma(l_2), \dots, \gamma(l_N))$ represents the sought position encoding for all pieces E
 The training losses for no missing pieces and with missing pieces are following (we use a **diffusion vision transformer for both ϵ_θ and E_θ**):

$$\mathcal{L}(\theta) = \begin{cases} \mathbb{E}_{L_0 \sim q(L_0), \epsilon \sim \mathcal{N}(0, I), t, E^g} [\|\epsilon - \epsilon_\theta(L_t, t, E^g)\|_2^2] = \mathcal{L}^{PE} \\ \lambda \mathcal{L}^{PE} + (1 - \lambda) \mathbb{E}_{L_0 \sim q(L_0), \epsilon \sim \mathcal{N}(0, I), t, E^g} [\|E^m - E_\theta(L_t, t, E^g)\|_2^2] \end{cases}$$

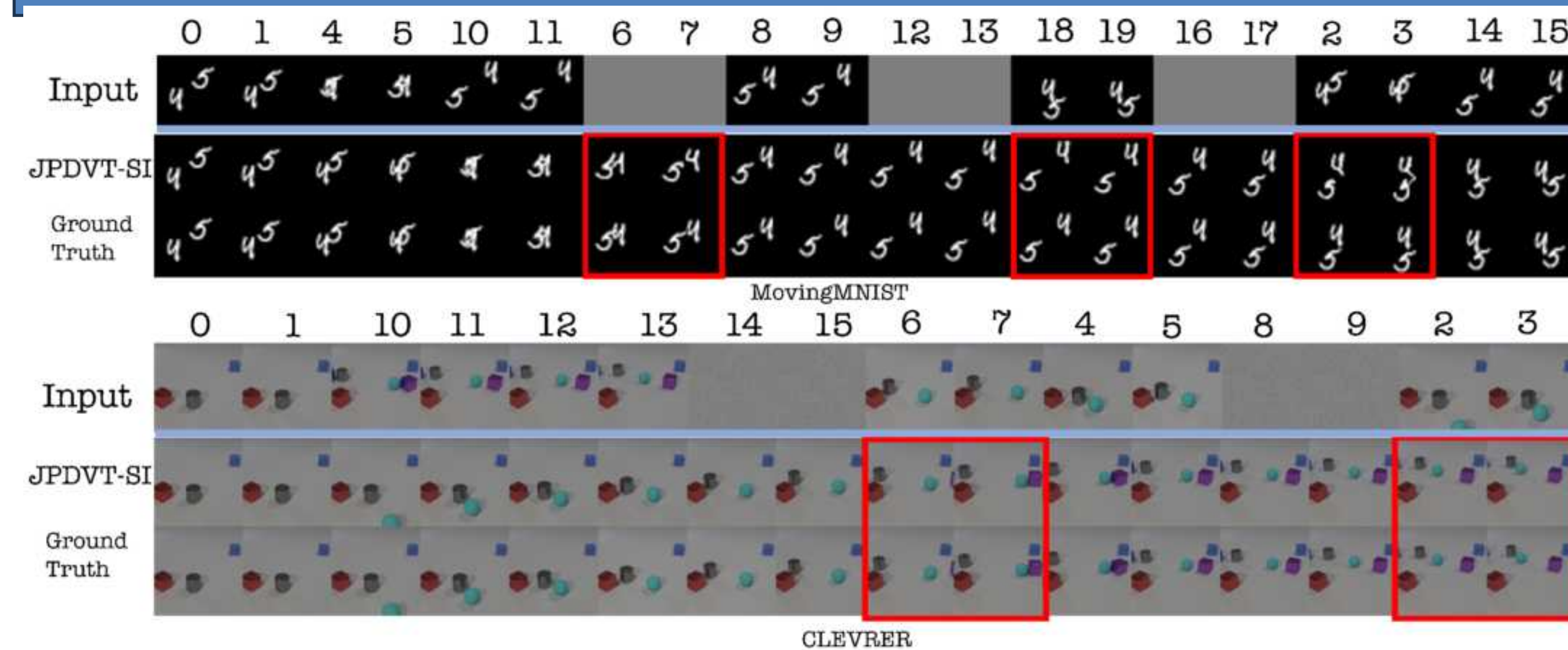
Spatial Results



Intuitive Insight



Temporal Results



Quantitative Results

3 × 3 Spatial Puzzles on *ImageNet*

Method	Puzzle-level	Piece-level
SAJ	47.3%	NA
Deepzle	48%	78%
JPDVT(Ours)	68.7%	83.3%

15 × 10 Spatial Puzzles

Method	Piece-level	Puzzle-level
SJP-ED	66.7%	10%
JPDVT(Ours)	75.9%	45%

3 × 3 Spatial Puzzles on *JPwLEG-3*

Method	Venue	Piece-level
Siamese	AAAI-2023	59.7%
Puzzlet	ICASSP-2023	58.2%
SJP-ED	CVPR-2020	55.2%
Deepzle	TIP-2020	52.3%
JPDVT(Ours)	CVPR-2024	71.3%

Temporal Puzzles

Datasets	Method	Normalized Kendall Distance x 10 ³			
Size, #Pieces		1, 20	2, 10	4, 5	
	STP	1.4	0.04	0.01	
	VCOP	-	-	13.7	
	JPDVT(Ours)	0.7	0.04	0	
MMNIST		1, 32	2, 16	4, 8	8, 4
	STP	-	0.15	0	0
	VCOP	-	-	-	4.6
	JPDVT(Ours)	2.6	0.17	0.04	0
CLEVER		1, 268.2	58.2	0.04	0
	VCOP	-	-	-	14.5
	JPDVT(Ours)	26.2	4.2	1.2	0.3
	STP	-	0	0	0
UCF		1, 164.3	3.8	0.5	
	VCOP	-	-	-	14.5
	JPDVT(Ours)	7.0	3.0	0.19	0.04
	STP	-	-	-	14.5





Solving Masked Jigsaw Puzzles with Diffusion Vision Transformers

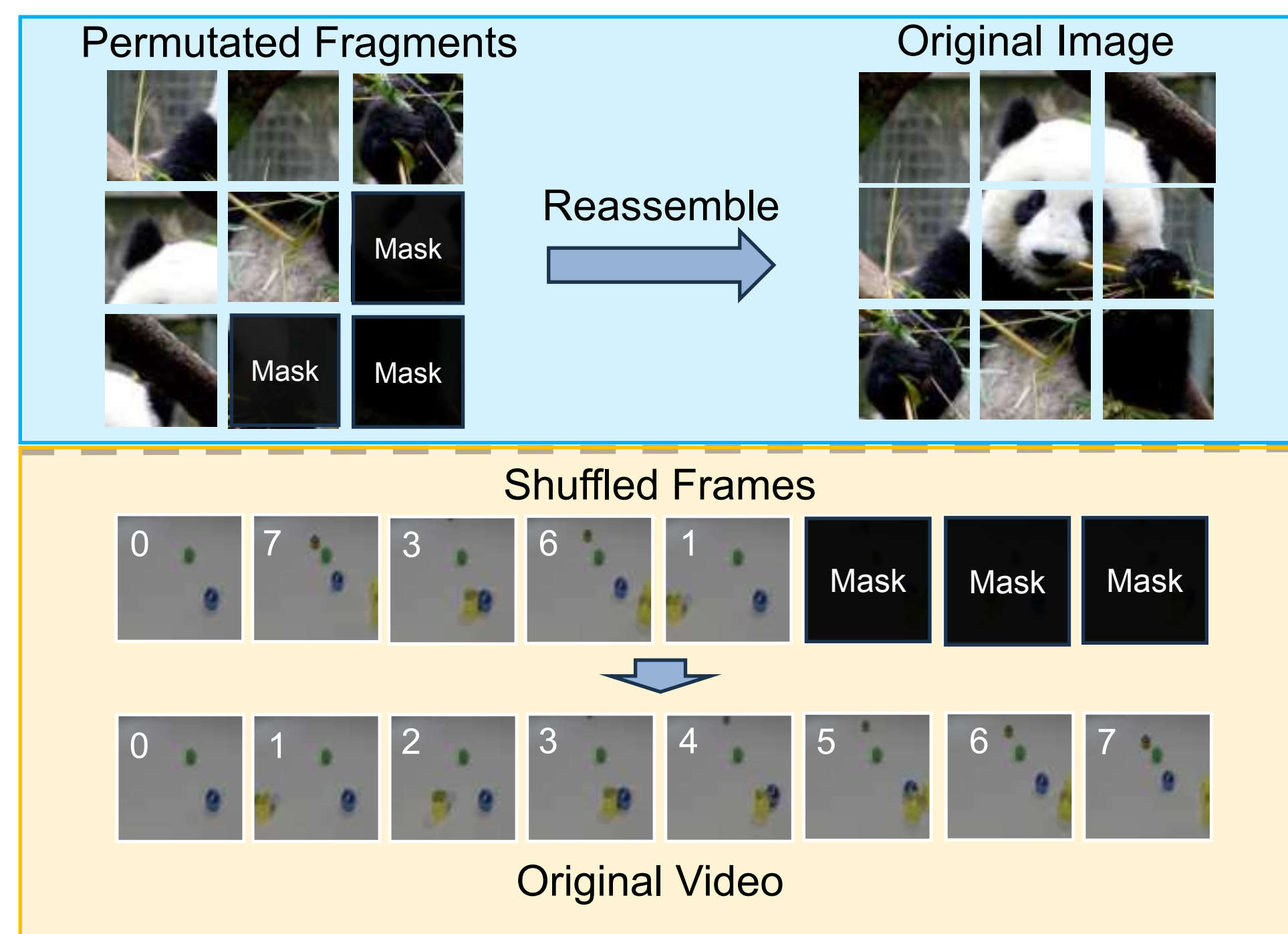
Jinyang Liu¹, Wondmgezahu Teshome¹, Sandesh Ghimire², Mario Sznai¹, Octavia Camps¹
1. Electrical & Computer Engineering Department, Northeastern University 2. Qualcomm Inc.



Masked Jigsaw Puzzles

Spatial

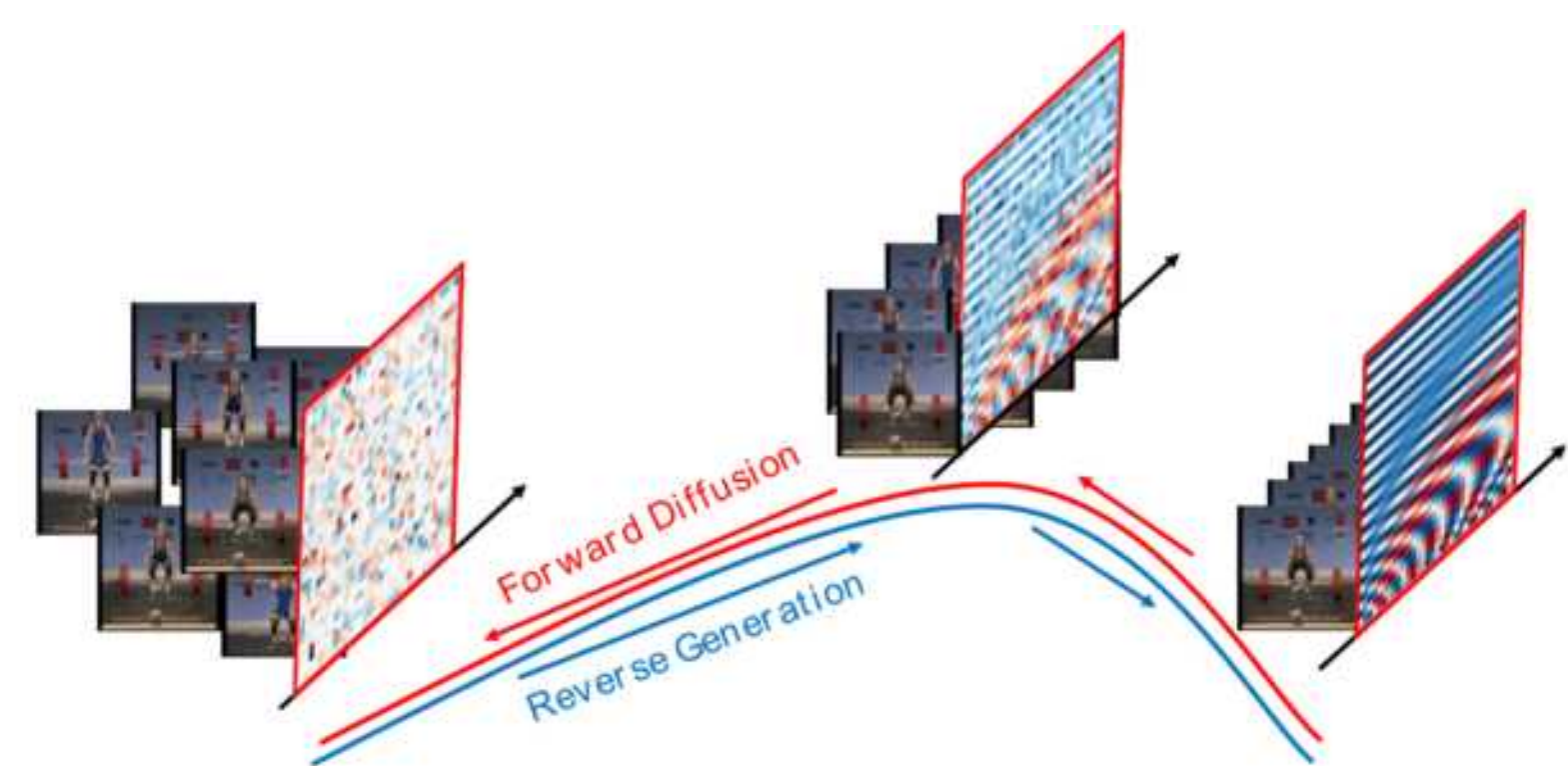
Temporal



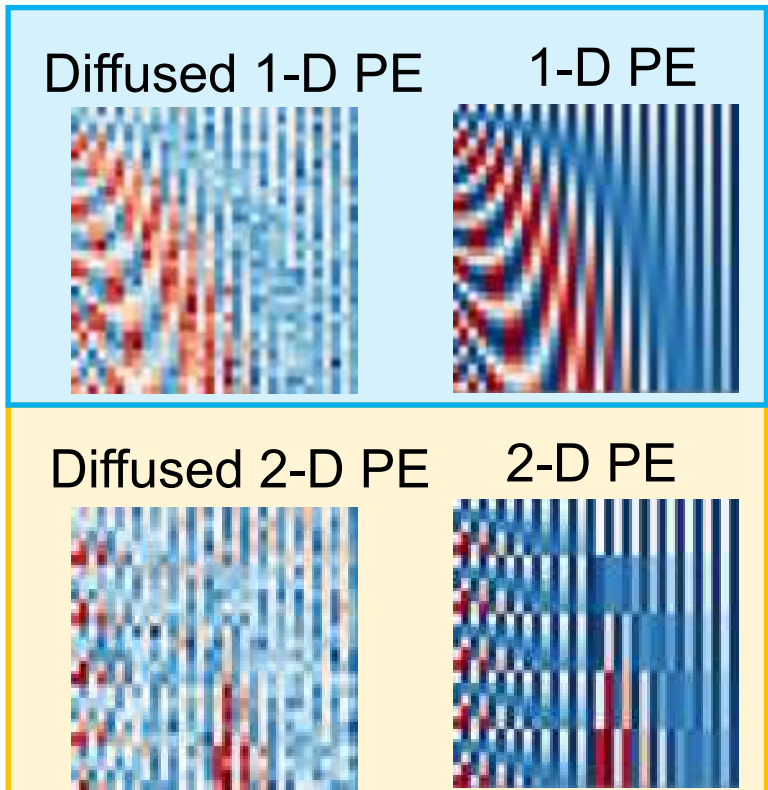
Masked Jigsaw Puzzle Solving Task:

Given an unordered sequence with missing data, reassemble and render the missing image fragments or video frames into meaningful sequences.

Intuitive Insight: Recover the Ordering

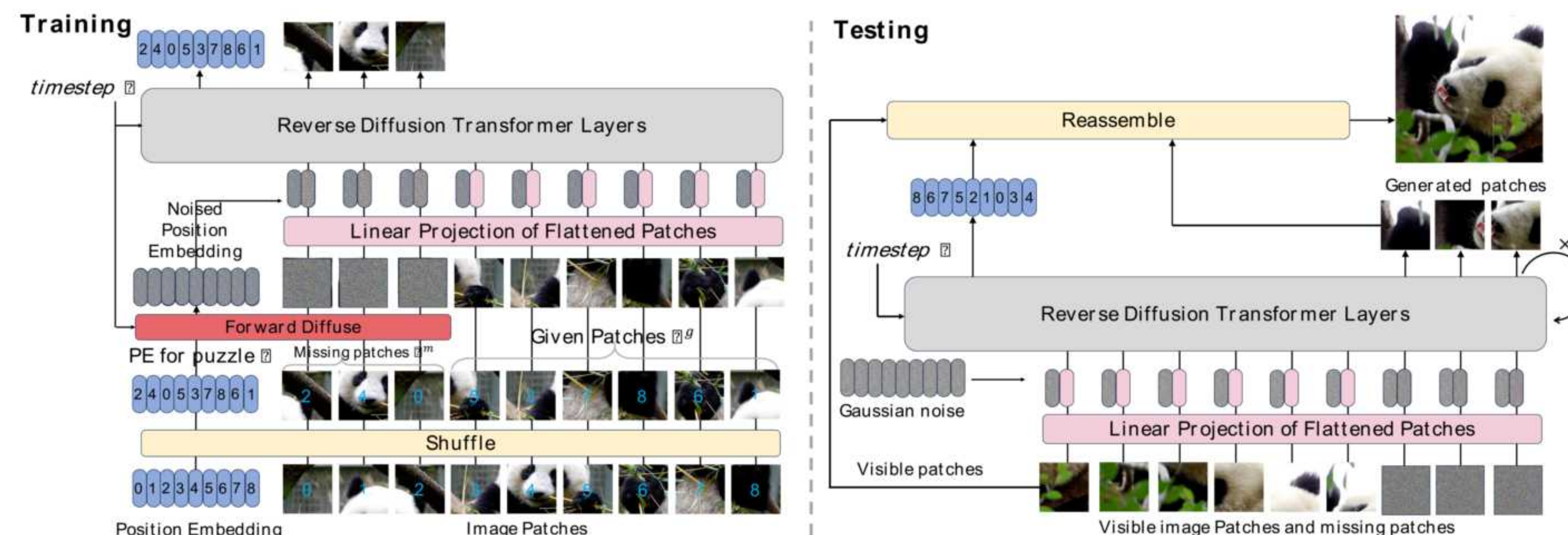


Positional Embedding Ex.



Distinct from previous discriminative models, we propose the use of a diffusion vision transformer to **generate the positional encoding of the fragments and missing pieces, conditioned on the given visible content.**

Our Method: JPDVT



$$\mathbf{E} = \{e_1, e_2, \dots, e_N\} = \mathbf{E}^g \cup \mathbf{E}^m: \text{given and missing pieces}$$

$$\mathbf{L} = (\gamma(l_1), \gamma(l_2), \dots, \gamma(l_N)): \text{sought position encodings for all pieces } \mathbf{E}$$

Training losses without and with missing pieces are:

(we use a **diffusion vision transformer for both ϵ_θ and $\mathbf{E}_\theta$**):

$$\mathcal{L}(\theta) = \begin{cases} \mathbb{E}_{\mathbf{L}_0 \sim q(\mathbf{L}_0), \epsilon \sim \mathcal{N}(0, I), t, \mathbf{E}^g} [\|\epsilon - \epsilon_\theta(\mathbf{L}_t, t, \mathbf{E}^g)\|_2^2] = \mathcal{L}^{PE} \\ \lambda \mathcal{L}^{PE} + (1 - \lambda) \mathbb{E}_{\mathbf{L}_0 \sim q(\mathbf{L}_0), \epsilon \sim \mathcal{N}(0, I), t, \mathbf{E}^g} [\|\mathbf{E}^m - \mathbf{E}_\theta(\mathbf{L}_t, t, \mathbf{E}^g)\|_2^2] \end{cases}$$

Quantitative Results

3 × 3 Spatial Puzzles on *ImageNet*

Method	Puzzle-level	Piece-level
SAJ	47.3%	NA
Deepzle	48%	78%
JPDVT(Ours)	68.7%	83.3%

3 × 3 Spatial Puzzles on JpWLEG-3

Method	Venue	Piece-level
Siamese	AAAI-2023	59.7%
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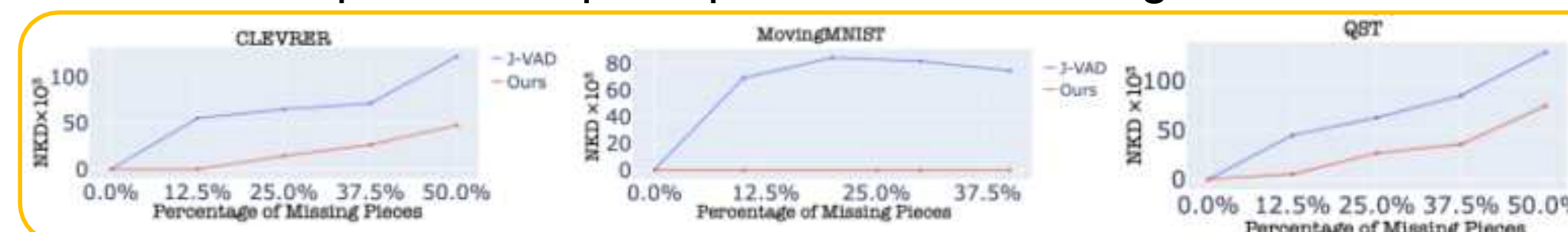
3 × 3 Spatial Puzzles on *ImageNet* with missing fragments

% Missing	With gap		Without gap	
	puzzle	piece	puzzle	piece
0	68.7%	83.3%	99.4%	99.6%
11.1	41.5%	72.0%	96.9%	98.8%
22.2	21.4%	61.8%	82.8%	95.0%
33.3	14.9%	54.1%	56.1%	76.4%

15 × 10 Large Spatial Puzzles

Method	Piece-level	Puzzle-level
SJP-ED	66.7%	10%
JPDVT(Ours)	75.9%	45%

8 pieces temporal puzzles with missing Pieces



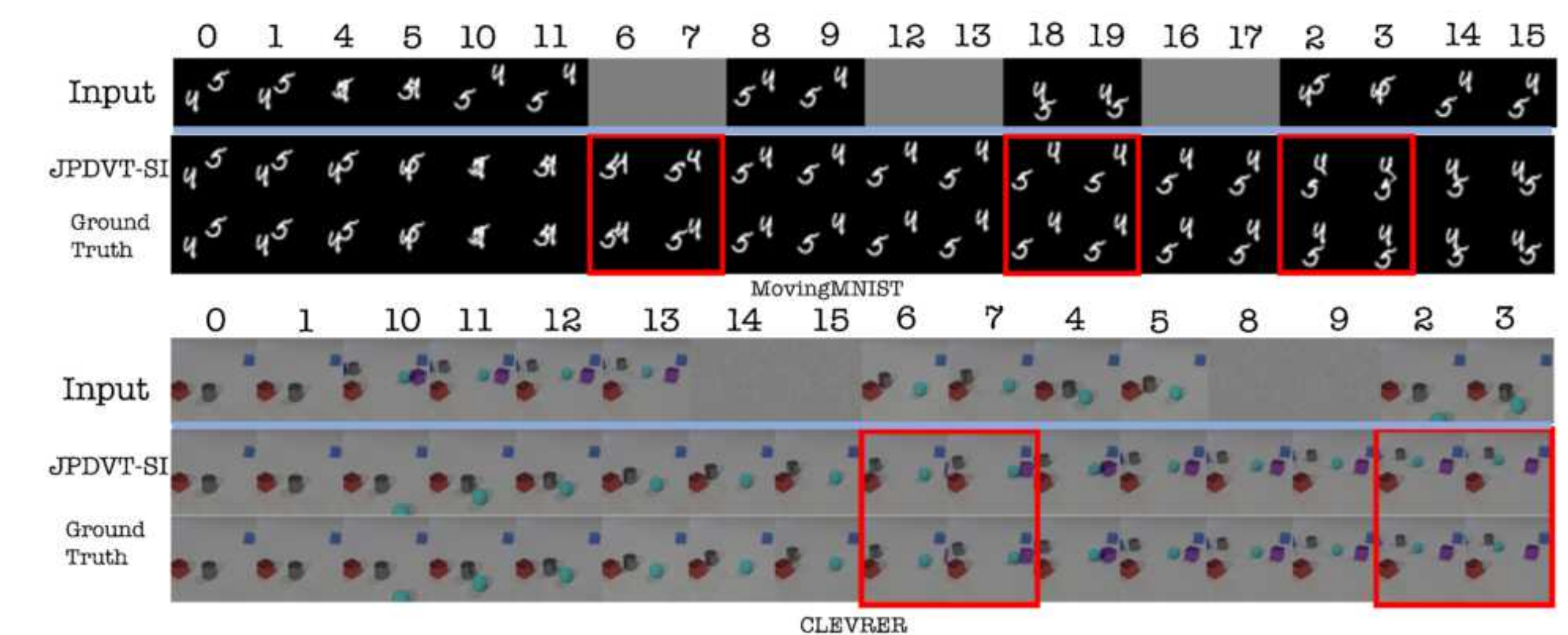
Temporal Puzzles

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	VCOP	1.4	0.04	0.01	
	J-VAD	-	-	13.7	
	JPDVT(Ours)	0.7	0.04	0	
MMNIST	STP	1, 32	2, 16	4, 8	8, 4
	VCOP	-	0.15	0	0
	J-VAD	173	0.46	0	
	JPDVT(Ours)	2.6	0.17	0.04	0
CLEVRER	STP	-	-	-	14.5
	VCOP	-	-	-	4.6
	J-VAD	268.2	58.2	0.04	0
	JPDVT(Ours)	26.2	4.2	1.2	0.3
UCF	STP	-	-	-	14.5
	VCOP	-	-	-	0
	J-VAD	-	164.3	3.8	0.5
	JPDVT(Ours)	7.0	3.0	0.19	0.04

Spatial Qualitative Results



Temporal Qualitative Results



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